A reproducible candidate framework for projection-sensitive residual structure in SPARC rotation curves

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Abstract

Paper 1 found that externally reviewed structural disturbance is associated with larger low-acceleration residual scatter in SPARC. Paper 2 reversed the question and showed that fixed residual-shape features can recover those A/C labels better than chance, while remaining explicitly non-unique with respect to MOND-simple and empirical RAR-like baselines. This Paper 3 seed asks a narrower follow-up question: where do TPG/projection residuals, MOND-simple residuals, RAR-like residuals, and measured rotation curves point to reproducible projection-sensitive residual candidates? The current packet identifies candidate galaxies, radial onset categories, environment/observability stress channels, and a frozen candidate/control ladder anchored by DDO126 and DDO50. It is a positive candidate-support methodology and registry paper: it turns the Paper 1–2 residual association into a concrete, auditable target list for the next predictive test. It does not validate a physical theory, does not numerically test RMOND, and does not claim gravity-model selection.

1 Purpose and claim boundary

The working hypothesis is that the TPG/projection baseline may capture part of a local projection-sensitive residual structure, while the remaining TPG residual may carry missing observer- and environment-dependent structure. In this manuscript, "TPG/projection" denotes a frozen operational projection baseline used for residual comparison rather than a validated gravity model; TPG is retained as a historical operational label from earlier work and does not imply a validated gravity theory. This is a candidate-search statement, not a validation result. A defensible Paper 3 must therefore separate three layers:

1. an operational residual layer, based on frozen TPG/projection, MOND-simple, and RAR-like residual maps;

2. a systematics layer, based on distance, inclination, point count, H I kinematics, beam smearing, and non-circular motions;

3. an interpretation layer, where any projection-sensitive residual interpretation is allowed only if the residual pattern survives the systematics layer and is more specific than ordinary RAR/MOND behavior.

The permitted claim is that this packet defines a reproducible candidate-selection and control framework for projection-sensitive residual structures. The forbidden claim is that the candidates already validate any physical theory or gravity model.

2 Data inherited from Papers 1 and 2

The packet inherits derived SPARC residual artifacts from the Paper 1 and Paper 2 public packages. Raw SPARC rotmod files and raw H I survey products are not redistributed. The retained derived tables include pointwise absolute residuals for TPG/projection, MOND-simple, and empirical RAR-like families; galaxy-level residual features; distance, observability, and environment summaries; and small radial-pilot control summaries.

The full Paper 3 reproducibility package, including the candidate/control tables, figures, regeneration script, compiled PDF, and arXiv source package, is archived at `doi:10.5281/zenodo.20285908`. The package can be regenerated with the commands listed in the repository README.

The main residual quantity is

$$r_{m,gi} = |\log V_{\mathrm{obs},gi} - \log V_{m,gi}|\,,$$

with galaxy-level burden

$$\mathrm{RMS}_{m,g} = \sqrt{\frac{1}{N_g}\sum_i r_{m,gi}^2}.$$

The TPG-specific excess used for candidate triage is

$$\Delta_{\mathrm{TPG-low},g} = \mathrm{RMS}_{\mathrm{TPG},g} - \min\left(\mathrm{RMS}_{\mathrm{MOND},g}, \mathrm{RMS}_{\mathrm{RAR},g}\right).$$

Positive values mark galaxies where the TPG/projection residual is larger than the best available low-acceleration comparator in this packet. Negative values mark TPG success controls.

3 Theory motivation

The relevant external baseline is the SPARC radial acceleration relation [4, 2] and the broader MOND/RAR literature [5, 3]. These works are not optional background; they define the strongest ordinary low-acceleration competitor. If a projection-sensitive residual hypothesis only reproduces the same residual ordering as RAR/MOND, it is not yet distinguished.

The systematics literature is equally central. Non-circular motions in H I velocity fields [8, 7] and beam-smearing/rotation-curve quality effects [1] can generate residual structure without new physics. The JWST/NIRCam Zone-of-Avoidance result [6] is used only as an observer/line-of-sight motivation: foreground obscuration and hidden structure can materially affect what an observer can map. It is not evidence for the projection-sensitive residual interpretation by itself.

4 Candidate construction

For each galaxy we record TPG/projection RMS, MOND-simple RMS, RAR-like RMS, TPG-specific excess, environment proxy, distance, reconstruction-risk proxy, inclination, and the first radial bin where the TPG/projection absolute residual exceeds 0.15 dex. Candidate classes are assigned by frozen screening rules:

- *clean projection-sensitive candidate*: high TPG residual, high TPG-specific excess, and non-high reconstruction risk;
- *environment-linked projection candidate*: high residual burden with high environment cue;

- *TPG divergence follow-up*: positive TPG-specific excess without enough cleanliness for the first two labels;
- *TPG success control*: low residual burden where TPG is no worse than MOND/RAR-like baselines.

These classes are triage labels. They are not physical classifications.

Table 1: Candidate shortlist from the regenerated seed packet.

Galaxy	Class	TPG RMS	TPG excess	Env.	Onset	Candidate class
UGC00731	C	0.340	0.093	0.00	inner_R<0.33Rmax	clean projection
CamB	A	0.964	-0.080	1.72	inner_R<0.33Rmax	environment linked
UGC05764	A	0.329	0.076	0.00	inner_R<0.33Rmax	TPG divergence
NGC2403	A	0.085	0.029	1.86	outer_R>=0.67Rmax	TPG divergence
NGC3741	C	0.204	0.103	-0.12	inner_R<0.33Rmax	TPG divergence
UGC08490	C	0.186	0.055	0.00	inner_R<0.33Rmax	TPG divergence
NGC3109	C	0.176	0.039	0.20	middle_0.33-0.67Rmax	TPG divergence
UGC06446	C	0.194	0.038	0.00	inner_R<0.33Rmax	TPG divergence

5 Residual and environment readouts

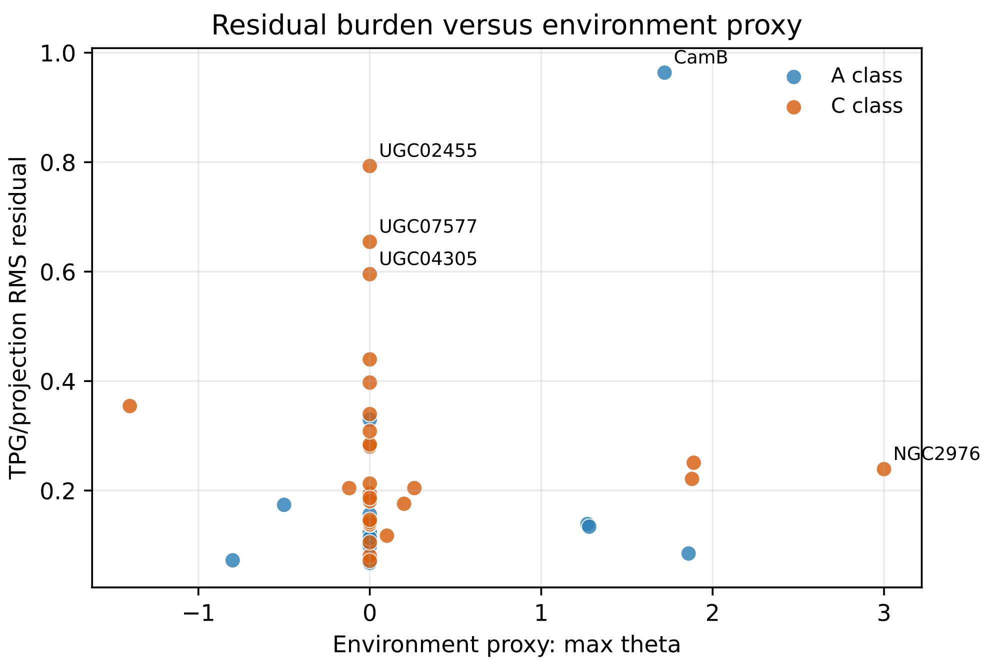

Figure 1: TPG/projection residual burden versus the inherited environment proxy. Named points are the largest seed-candidate scores.

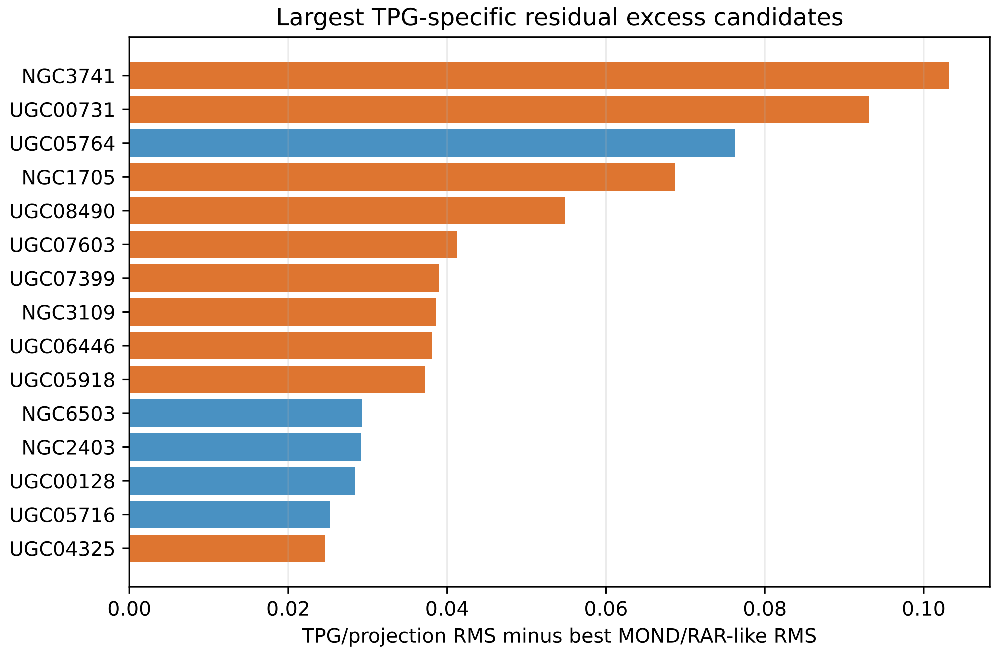

Figure 2: Galaxies with the largest TPG/projection residual excess relative to the better of MOND-simple and RAR-like RMS.

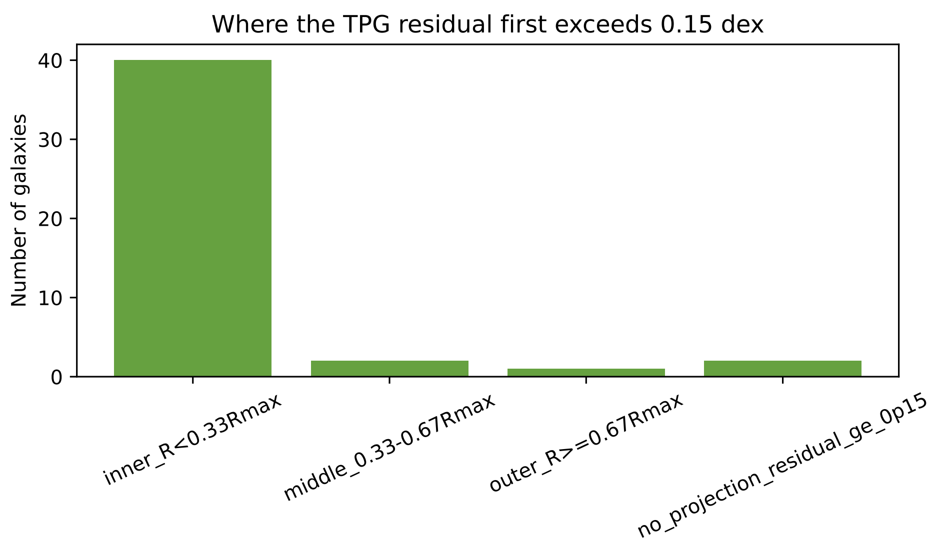

Figure 3: Radial bin where the TPG/projection absolute residual first exceeds 0.15 dex. This is a morphology-of-failure diagnostic, not a detection statistic.

Table 2: Screening correlations for TPG/projection residual burden.

Covariate	Pearson r	Interpretation
DistanceMpc	-0.291	screening_correlation_not_causal_attribution
EnvMaxTheta	0.110	screening_correlation_not_causal_attribution
W_tau_eff_abs_v01	0.983	screening_correlation_not_causal_attribution
ReconstructionRiskChannel_v01	0.317	screening_correlation_not_causal_attribution
MeanErrVobsKms	-0.214	screening_correlation_not_causal_attribution
InclinationErrorDeg	0.214	screening_correlation_not_causal_attribution

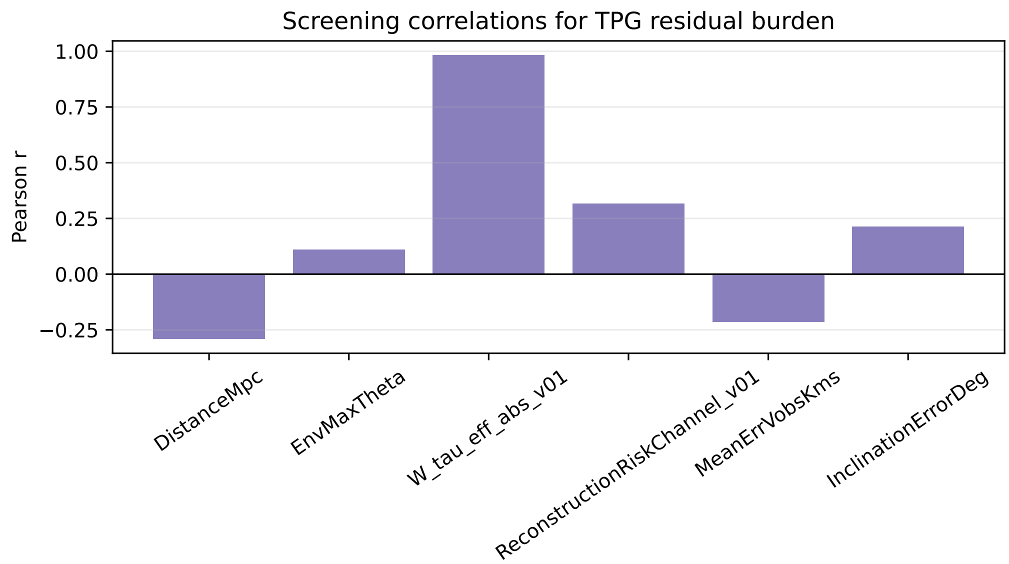

Figure 4: Screening correlations with the TPG/projection residual burden. These are not causal estimates.

One covariate in this table, W_tau_eff_abs_v01, is intentionally treated as an internal diagnostic rather than independent evidence. Its correlation with projection residual burden is very high, and the variable is not allowed to support any validation claim until a leakage audit proves that it is not partly defined from the same residual endpoint. In the present manuscript it is therefore a bookkeeping stress term only.

6 Endpoint-conditioned S_τ diagnostic

The descriptive projection-sensitive residual form can be written operationally as

$$F_\tau(a_N, R) = 1 + S_\tau(R)\,\alpha \ln\left(1 + \frac{a_0}{a_N}\right).$$

In this section we do not fit a new model. We algebraically invert this expression after substituting the observed-to-baryonic velocity ratio at each radius. For each SPARC point with a usable baryonic baseline, the endpoint-conditioned descriptive value is therefore

$$S_{\tau,\mathrm{req}}(R) = \frac{V_{\mathrm{obs}}(R)/V_N(R) - 1}{\alpha \ln\left(1 + a_0/a_N(R)\right)}.$$

Here $S_{\tau,\mathrm{req}}$ is the value that would be required for the operational form above to absorb the local residual. This is an inverse residual-absorption diagnostic. It is descriptive and endpoint-conditioned. It uses the measured endpoint and therefore is not a predictive field reconstruction, not independent evidence for a physical theory, and not a validation of the projection ansatz. Its role is only to ask what kind of future predictive rule would be needed: a galaxy-level constant, a radial function, an acceleration function, or an environment-coupled function.

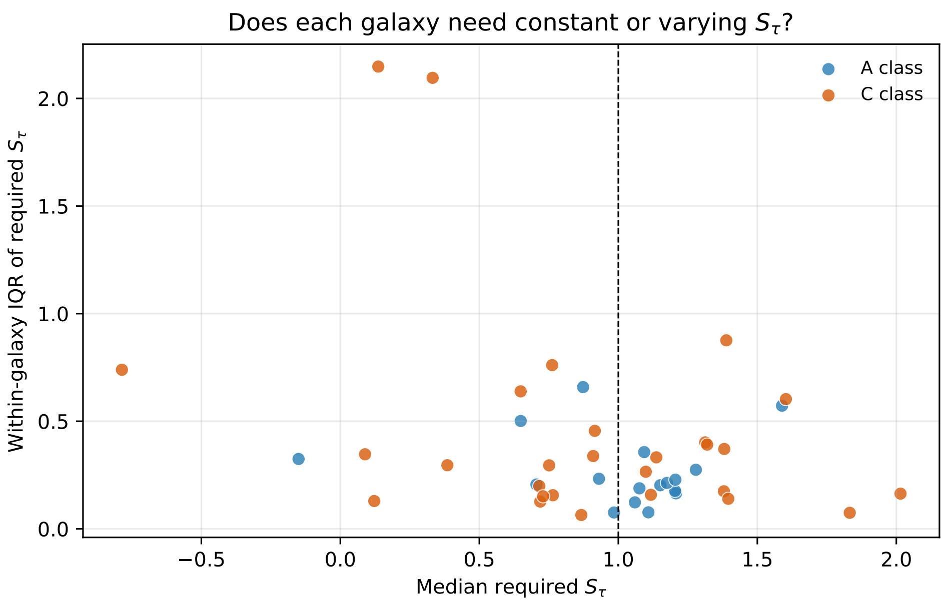

Figure 5: Median required S_τ versus within-galaxy spread. A low spread near $S_\tau = 1$ would support a constant correction; large spread indicates that a function is needed.

Table 3: In-sample required-S_τ function-family diagnostics. Lower RMS means that family can absorb the measured TPG residual more compactly.

Family	Median RMS	Mean RMS	Galaxies
radius + acceleration	0.143	0.152	45
quadratic radius	0.145	0.153	45
linear radius	0.146	0.168	45
quadratic acceleration	0.165	0.160	45
linear acceleration	0.165	0.167	45
galaxy constant	0.187	0.183	45
fixed S=1	0.188	0.249	45

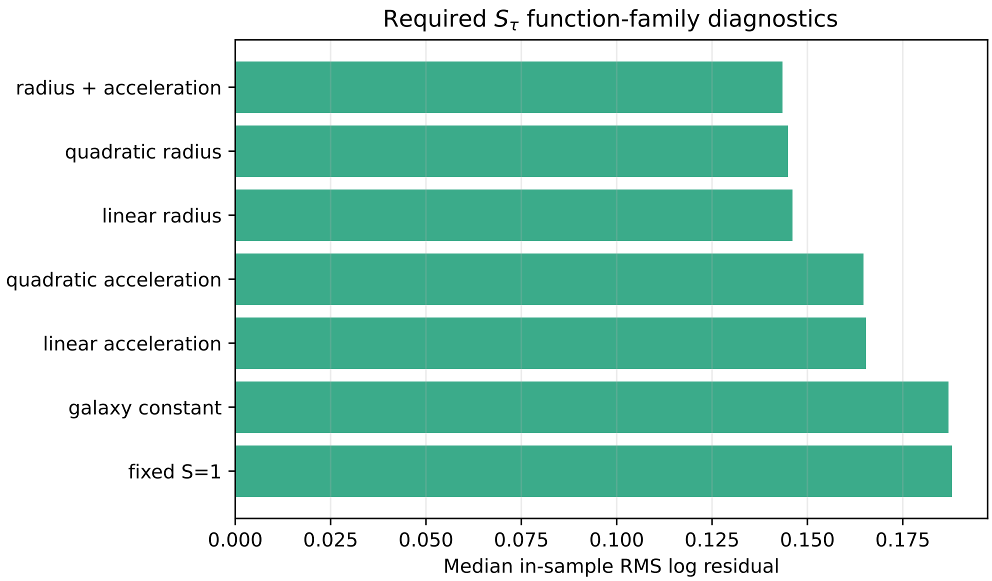

Figure 6: Diagnostic comparison of constant and simple functional S_τ families. This is not model selection because the measured velocities are used to infer S_τ.

7 Small-residual controls

The same diagnostic can be read from the opposite direction. The lowest-quartile TPG/projection RMS objects form a small-residual control set of 12 galaxies, with median projection RMS 0.099854397 and median required $S_\tau = 1.126924435$. These objects ask why the TPG/projection baseline can remain close to the measured curve, rather than why it fails.

The audit gives a cautious answer. One object is classified as a possible local-capture case, where the local TPG/projection baseline may already absorb the dominant local projection-sensitive structure. Three objects are better described as cases where the broader low-acceleration family tracks the measurement well. The remaining eight small-residual objects still carry unresolved radial or external residual structure. Thus good TPG agreement is useful control context, not validation. These rows must remain paired with the high-residual candidates in any future source-native endpoint test.

8 Heuristic residual-pattern decomposition

The DDO75/Sextans A stress case motivates a compact heuristic decomposition of the unknown residual pattern. This subsection is not a model-building result and is not used as evidence in the present packet. It is included only to state what a future nonleaky predictor might have to separate. In the working form above, the logarithmic low-acceleration kernel is universal, while the remaining factor $S_\tau(R)$ is allowed to carry source and observer dependence:

$$S_\tau(R) = S_0 + \beta_{\rm src} C_{\rm src}(R) + \beta_{\rm path} C_{\rm path}(D) + \beta_{\rm proj} C_{\rm proj}(B, R_{\rm max}) + \beta_{\rm drift} C_{\rm drift}(R).$$

Here $C_{\rm src}$ is source-side radial structure, $C_{\rm path}$ is observer path length or distance, $C_{\rm proj}$ is projection/resolution, and $C_{\rm drift}$ is radial-drift or outer-curve morphology. All four terms are placeholders for future preregistered predictors. None is allowed to use target residuals, required S_τ, or post-hoc endpoint gains as an input.

9 RMOND comparator status

RMOND is retained only as a roadmap comparator. The current packet does not contain a frozen pointwise $V_{\rm RMOND}(R)$ law, and therefore it does not contain an RMOND residual endpoint. Local bridge notes show theory-level compatibility and also warn about double counting: at $f_{\rm vec} = 0$ the hybrid reduces to the TPG/MTW case, while nonzero vector terms require an independently frozen halo/vector prescription. A future version may add RMOND only after a unique velocity law is preregistered and evaluated on the same SPARC radii as TPG/projection, MOND-simple, and RAR-like baselines.

10 Post-repair status and next gate

After the initial seed construction, one repaired branch, DDO168, was rescored under frozen rules as a ladder-consistency example rather than as a primary endpoint. The repaired DDO168 input layer removes the previous mass-closure caveat and gives a single-object specificity-support result: empirical RAR-like remains the best low-acceleration score, but fixed TPG lies within the frozen tie tolerance of the best low-acceleration baseline, with $\Delta_{\rm TPG-best} = 0.001509618$ in RMS-log units. The Newtonian baryonic score is worse than fixed TPG by 0.089064802, and the inverse required-S_τ diagnostic remains strongly radial, with Pearson $r = 0.927628669$.

This result is useful because it places DDO168 beside visible anchors and controls rather than treating it as an isolated success. DDO126 is a positive fixed-TPG public endpoint, DDO50 is a quiet Newtonian-best control with near-zero required-S_τ, DDO154 is a near-one control/countercontrol, and NGC2366 is a caveated lower-amplitude radial candidate. DDO168 therefore strengthens the candidate ladder as a specificity-support example. It is not validation evidence by itself, and it does not select TPG over RAR/MOND as a gravity model.

The next gate is therefore two-track. The paper can now be written as a narrow status or seed paper whose claim is candidate specificity support in context. Any stronger claim requires the already frozen eight-object clean endpoint expansion: DDO154, DDO168, NGC2366, WLM, IC1613, DDO47, DDO50, and DDO126, with auditable component-velocity inputs or frozen reconstructions before running the LOGO TPG/MOND/RAR/radial-S_τ endpoint.

11 Post-external-support status

The later external-support branch adds a useful but deliberately limited result. A preregistered external-only screen is directionally positive: the candidate side has larger endpoint burden than the mandatory controls and countercases under the frozen readout. This supports the idea that the residual candidates are not purely random bookkeeping artifacts. However, the same branch remains observability-caveated. Distance, inclination, point count, H I resolution, and source visibility remain possible ordinary explanations, and the external branch cannot substitute for accepted baryonic component data.

For this reason, the current Paper 3 claim is bounded as follows. The manuscript may report a coherent candidate-support pattern in residual and external-proxy diagnostics. It may also report that a future external-plus-component readout has been frozen as a nonexecuted contract. It may not claim that a physical theory has been validated, that TPG has been selected over MOND/RAR, or that the missing source-native component endpoint has been satisfied.

The frozen combined contract compares three future predictors only after accepted component inputs pass intake: an external-only predictor, a component-only predictor, and an external-plus-component predictor. Endpoint outputs and diagnostics such as Vobs residuals, FunctionGain, ProjectionRMS_TPG, and required S_τ are forbidden as predictor inputs. This keeps the validation route alive without allowing endpoint-driven retuning.

12 Anchor/control rotation curves

Figure 7 shows the two most important current visual endpoints. DDO126 is the positive public-only anchor candidate, and DDO50 is the required control. The panels are diagnostic visualizations of the frozen candidate/control ladder. They are not independent validation and they do not replace the component-input gate.

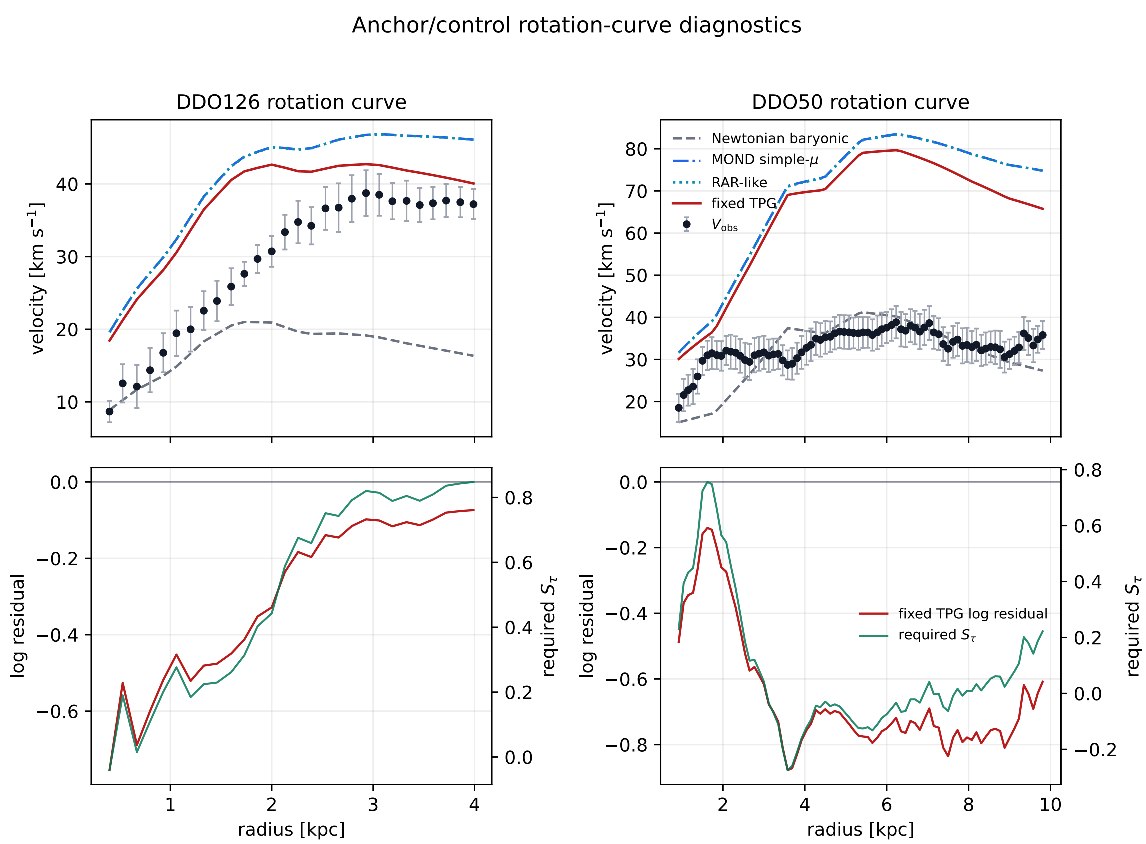

Figure 7: Anchor/control rotation-curve diagnostics. Top panels compare observed rotation speeds with the frozen Newtonian baryonic, MOND-simple, empirical RAR-like, and fixed-TPG curves. Bottom panels show the fixed-TPG log residual and the inverse required-S_τ diagnostic. The figure is a visual check on the candidate/control ladder, not a validation endpoint.

13 Current candidate-ladder status

The later DDO126/DDO50 and held-out readiness audits sharpen the current claim boundary. The manuscript can report a reproducible candidate ladder with named positive anchors, mandatory controls, and failure-mode objects. DDO126 is the strongest public-only positive anchor candidate in the present packet, while DDO50 is the required control branch. WLM is retained as a geometry/observability stress context. NGC2366 remains caveated radial support and must be reported with DDO154 as its countercontrol. IC1613 and DDO47 remain sensitivity and failure-mode objects until their stellar or component-input blockers are repaired.

This status does not add a new endpoint score. It is a claim-control layer. The input watchlist defines what would be needed for a future upgrade: schema-clean component intake for DDO126/DDO50, or a broader component-confirmed held-out pool. Any new file must pass provenance, column-schema, unit/radius, geometry/systematics, and control-visibility checks before model scores are rerun.

Therefore the current paper-level claim is bounded but nontrivial. The packet establishes a reproducible projection-sensitive candidate/control ladder and freezes the component-intake route required to test it. The result is not a physical validation, not gravity-model selection, and not evidence that the residual pattern is unique to a projection-sensitive interpretation. Additional scoring without new accepted component or stellar inputs is not a claim-upgrade path.

14 Predictive validation gate

The present manuscript is not a validation paper, but it does more than list curiosities. It defines where a projection-sensitive residual signal should be looked for, which controls must travel with the candidates, and what predictor inputs are forbidden. It does not yet supply a predictive residual-pattern rule. A validation-oriented version must pass a stricter gate before any discovery language is appropriate.

First, the residual-pattern rule must be frozen before reading the endpoint residuals. The rule may use source-side structure, distance or line-of-sight information, projection/observability quantities, and preregistered radial-shape information, but it may not use Vobs residuals, FunctionGain, ProjectionRMS_TPG, or the inverse required-S_τ diagnostic as predictor inputs. Second, the rule must be evaluated on a predeclared clean candidate/control set, including both positive anchors and controls. Third, the result must improve the frozen endpoint relative to TPG/projection, MOND-simple, and empirical RAR-like baselines, not merely reproduce a generic low-acceleration ordering. Fourth, the improvement must survive the ordinary observability nulls: distance, inclination, point count, beam smearing, H I resolution, and non-circular motion.

This predictive gate is the main upgrade path from a seed paper to a validation paper. In practical terms, the next claim-raising experiment is: freeze a predictive projection-sensitive residual rule, then test whether it improves the predeclared clean candidate/control set beyond TPG/projection, MOND-simple, and RAR-like baselines. Until that experiment is passed, the correct status is candidate-support methodology rather than physical evidence.

15 Interpretation

The most useful interpretation is modest. TPG success controls show where the local projection baseline is already adequate. TPG divergence follow-ups show where the residual has structure that might encode missing observer/environment-sensitive information, but these are exactly the cases where ordinary systematics can also enter. A candidate becomes interesting only when three facts hold together: the TPG residual diverges in a structured radial way, the divergence is not shared by all low-acceleration baselines, and the object has a plausible projection-sensitive residual pattern that is not reducible to distance, inclination, beam smearing, or non-circular motion.

16 Conclusion

This paper establishes a reproducible candidate-search framework rather than merely proposing one. It carries forward the discipline learned from Papers 1 and 2: freeze the endpoint, compare against MOND/RAR baselines, name the systematics, and avoid promoting a diagnostic pattern into physical evidence. The repaired DDO168 branch, the external-support branch, and the DDO126/DDO50 candidate-ladder lock strengthen the candidate-specificity story in context: the residual candidates are now organized into a falsifiable target set with controls and blockers. The current evidence is still not enough to claim a detected physical field.

The next paper-grade step is the eight-object clean endpoint expansion, with source-native component tables or frozen public reconstructions wherever source-native tables are unavailable. The immediate action is therefore write-or-wait: update the manuscript with candidate-support language, or wait for source-native component input and run the intake gate. A frozen RMOND residual table and a held-out environment/line-of-sight validation rule remain important parallel upgrades. Only after those gates can the paper ask whether the residual candidates are truly projection-sensitive rather than ordinary low-acceleration or H I-systematics behavior.

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